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Safety data sheet

BASF Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)
Date / Revised: 29.12.2023 Version: 3.0
Product: **Assist EC**
(ID no. 30686580/SDS_GEN_00/EN)
Date of print 03.07.2025

1. Identification

Product identifier

Assist EC

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: adjuvant

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Crop Protection

Telephone: +49 621 60-27777
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2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Asp. Tox. 2
Skin Corr./Irrit. 3
Eye Dam./Irrit. 2A
Aquatic Acute 2

Aquatic Chronic 3

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Warning

Hazard Statement:

H319	Causes serious eye irritation.
H316	Causes mild skin irritation.
H305	May be harmful if swallowed and enters airways.
H412	Harmful to aquatic life with long lasting effects.
H401	Toxic to aquatic life.

Precautionary Statement:

P101	If medical advice is needed, have product container or label at hand.
P102	Keep out of reach of children.
P103	Read carefully and follow all instructions.

Precautionary Statements (Prevention):

P280	Wear eye protection.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P301 + P310	IF SWALLOWED: Immediately call a POISON CENTER or physician.
P337 + P313	If eye irritation persists: Get medical attention.
P331	Do NOT induce vomiting.
P332 + P313	If skin irritation occurs: Get medical attention.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

According to UN GHS criteria

Hazard determining component(s) for labelling: Lubricating oils

Other hazards

According to UN GHS criteria

See section 12 - Results of PBT and vPvB assessment.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

blend, Emulsifiable concentrate (EC)

Hazardous ingredients (GHS)
According to UN GHS criteria

Lubricating oils

Content (W/W): < 95 %	Asp. Tox. 2
CAS Number: 74869-22-0	H305
EC-Number: 278-012-2	
INDEX-Number: 649-484-00-0	

Poly(oxy-1,2-ethanediyl), .alpha.-(2-propylheptyl)-.omega.-hydroxy-

Content (W/W): < 5 %	Acute Tox. 5 (oral)
CAS Number: 160875-66-1	Skin Corr./Irrit. 3
	Eye Dam./Irrit. 2B
	Aquatic Acute 2

solvent naphtha

Content (W/W): < 5 %	Asp. Tox. 1
CAS Number: 64742-94-5	Flam. Liq. 4
	Skin Corr./Irrit. 2
	STOT SE 3 (drowsiness and dizziness)
	Aquatic Acute 2
	Aquatic Chronic 2
	H227, H315, H304, H336, H401, H411

Alcohols, C12-14, ethoxylated (> 1 < 2.5 mol EO)

Content (W/W): < 5 %	Aquatic Acute 1
CAS Number: 68439-50-9	Aquatic Chronic 2
	M-factor acute: 10
	H411, H400

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts

Content (W/W): < 5 %	Acute Tox. 4 (dermal)
CAS Number: 68953-96-8	Skin Corr./Irrit. 2
EC-Number: 273-234-6	Eye Dam./Irrit. 1
	Aquatic Acute 3
	Aquatic Chronic 2
	H318, H315, H312, H402, H411

Alcohols, C11-14-iso-, C13-rich, ethoxylated

Content (W/W): < 3 %	Acute Tox. 4 (oral)
CAS Number: 78330-21-9	Eye Dam./Irrit. 1
	Aquatic Acute 1
	Aquatic Chronic 2
	H318, H302, H411, H400

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention. Do not induce vomiting due to aspiration hazard.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media
Suitable extinguishing media:
water spray, dry powder, foam, carbon dioxide

Special hazards arising from the substance or mixture
Carbon monoxide, Carbon dioxide, nitrogen oxides, sulfur oxides
The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters
Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Keep containers cool by spraying with water if exposed to fire. In case of fire and/or explosion do not breathe fumes. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures
Do not breathe vapour/spray. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

Environmental precautions
Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations.

7. Handling and Storage

Precautions for safe handling
No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
Vapours may form ignitable mixture with air. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Conditions for safe storage, including any incompatibilities
Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

64742-94-5: solvent naphtha
74869-22-0: Lubricating oils

Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for higher concentrations or long-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic and alkaline compounds (e.g. EN 14387 Type ABEK).

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	liquid
Colour:	No data available.
Odour:	odourless
Odour threshold:	not applicable, odour not perceivable
pH value:	The product has not been tested.

Melting point:	The product has not been tested.
Boiling point:	approx. 309 °C (760 mmHg) Information based on the main component/s.
Flash point:	approx. 184 °C Information based on the main component/s.
Evaporation rate:	not applicable
Flammability:	not applicable
Lower explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.
Upper explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.
Ignition temperature:	> 300 °C The product has not been tested. The statement has been derived from the properties of the individual components.
Vapour pressure:	approx. < 666,5 hPa (25 °C) The product has not been tested. The statement has been derived from the properties of the individual components.
Density:	approx. 0,85 g/cm3 (20 °C)
Relative vapour density (air):	not applicable
Solubility in water:	insoluble
Partitioning coefficient n-octanol/water (log Kow):	not applicable
Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.
Viscosity, kinematic:	13,8 mm2/s (ASTM D445) (40 °C) Information based on the main component/s.
Explosion hazard:	Based on the chemical structure there is no indication of explosive properties.
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.

Other information

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid
See SDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Poly(oxy-1,2-ethanediyl), .alpha.-(2-propylheptyl)-.omega.-hydroxy-

Experimental/calculated data:
LD50 rat (oral): > 2.000 - 5.000 mg/kg (OECD Guideline 423)
Literature data.

Information on: Poly(oxy-1,2-ethanediyl), .alpha.-(2-propylheptyl)-.omega.-hydroxy-
Experimental/calculated data:
LD50 rat (dermal): > 5.000 mg/kg (OECD Guideline 402)

Irritation

Assessment of irritating effects:
Eye contact causes irritation. Skin contact causes slight irritation. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts
Experimental/calculated data:
Skin corrosion/irritation rabbit: Irritant. (FHSA Guideline)

Information on: solvent naphtha
Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Skin corrosion/irritation rabbit: Irritant. (other)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: Poly(oxy-1,2-ethanediyl), .alpha.-(2-propylheptyl)-.omega.-hydroxy-
Experimental/calculated data:
Serious eye damage/irritation rabbit: Irritant. (OECD Guideline 405)
Literature data.

Information on: Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts
Experimental/calculated data:
Serious eye damage/irritation rabbit: irreversible damage (FHSA Guideline)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Analogous: Assessment derived from products with similar chemical character.

Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts
Experimental/calculated data:
Guinea pig maximization test guinea pig: Non-sensitizing. (OECD Guideline 406)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Analogous: Assessment derived from products with similar chemical character.

Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: solvent naphtha
Assessment of carcinogenicity:
Long-term exposure to highly irritating concentrations resulted in skin tumors in animals. A carcinogenic effect in humans can be excluded after brief skin contact. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts
Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The substance may cause damage to the kidney after repeated ingestion of high doses, as shown in animal studies.

Aspiration hazard

May also damage the lung at swallowing (aspiration hazard).
The product has not been tested. The statement has been derived from the properties of the individual components.

Other relevant toxicity information

Misuse can be harmful to health.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Toxic to aquatic life. Harmful to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Alcohols, C12-14, ethoxylated (> 1 < 2.5 mol EO)
Toxicity to fish:
LC50 (96 h) 0,423 mg/l, Pimephales promelas (OECD Guideline 203, semistatic)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: Alcohols, C12-14, ethoxylated (> 1 < 2.5 mol EO)
Aquatic invertebrates:
EC50 (48 h) 0,125 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: Alcohols, C12-14, ethoxylated (> 1 < 2.5 mol EO)
Aquatic plants:
EC50 (72 h) 0,0445 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
The statement of the toxic effect relates to the analytically determined concentration.

EC10 (72 h) 0,034 mg/l (growth rate), Desmodesmus subspicatus (Algal growth inhibition test, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: Alcohols, C12-14, ethoxylated (> 1 < 2.5 mol EO)
Chronic toxicity to fish:
EC10 (30 d) 0,251 mg/l, Pimephales promelas (calculated, Flow through.)

Information on: Alcohols, C12-14, ethoxylated (> 1 < 2.5 mol EO)
Chronic toxicity to aquatic invertebrates:
EC10 (21 d) 0,054 mg/l, Daphnia magna (calculated, other)

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Alcohols, C12-14, ethoxylated (> 1 < 2.5 mol EO)
Assessment biodegradation and elimination (H2O):

Readily biodegradable (according to OECD criteria).

Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Alcohols, C12-14, ethoxylated (> 1 < 2.5 mol EO)
Assessment bioaccumulation potential:
Does not accumulate in organisms.

Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Alcohols, C12-14, ethoxylated (> 1 < 2.5 mol EO)
Assessment transport between environmental compartments:
Volatility: The substance will slowly evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is possible.

Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in the Montreal Protocol on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for	None known

user

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Asp. Tox.	Aspiration hazard
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Acute Tox.	Acute toxicity
Flam. Liq.	Flammable liquids
STOT SE	Specific target organ toxicity — single exposure
H305	May be harmful if swallowed and enters airways.
H227	Combustible liquid.
H315	Causes skin irritation.
H304	May be fatal if swallowed and enters airways.
H336	May cause drowsiness or dizziness.
H401	Toxic to aquatic life.
H411	Toxic to aquatic life with long lasting effects.
H400	Very toxic to aquatic life.
H318	Causes serious eye damage.
H312	Harmful in contact with skin.
H402	Harmful to aquatic life.
H302	Harmful if swallowed.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.